

SAP Inventory Management System

Capstone Project Documentation

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Project Type:	Full-Stack Web Application
SAP Module:	MM — Materials Management

1. Problem Statement

In today's competitive manufacturing and supply chain environment, organizations heavily rely on ERP systems like SAP for managing their inventory, procurement, and vendor relationships. However, understanding and using these systems requires hands-on practice which is often not accessible to students and trainees due to the high cost and complexity of SAP licensing.

This project addresses the challenge of providing a practical, browser-based simulation of SAP's Materials Management (MM) module that covers the full procurement cycle — from creating material master records and onboarding vendors, to raising purchase orders, posting goods receipts, and monitoring live stock levels — without requiring any SAP installation or license.

2. Solution & Features

The SAP Inventory Management System (SAP IMS) is a complete web application that simulates real SAP MM workflows. It is built with pure HTML, CSS, and JavaScript — making it instantly deployable on any browser and hostable on GitHub Pages with zero backend configuration.

Module	SAP Transaction	Description
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Dashboard	—	Live KPI cards, trend charts, low-stock alerts, recent PO list
Material Master	MM01 / MM60	Full CRUD for raw, semi-finished and finished materials
Vendor Master	XK01 / XK03	Vendor onboarding with rating, payment terms, and status
Purchase Orders	ME21N / ME23N	Multi-line PO creation, approval workflow, status tracking
Goods Receipt	MIGO / MB01	Post GR against approved POs with automatic stock update
Stock Overview	MMBE / MB52	Real-time stock view, manual adjustment, stock valuation
Reports	—	Vendor analysis, stock valuation, PO summary charts

3. Tech Stack

Layer	Technology	Purpose
Structure	HTML5	Semantic page structure and multi-page routing
Styling	CSS3 / CSS Variables	SAP-themed design system with dark sidebar
Logic	Vanilla JavaScript ES6+	All business logic, CRUD, filtering, validation
Charts	Chart.js v4.4.0	Bar, doughnut, and pie chart visualizations
Icons	Font Awesome 6.4.0	SAP-style iconography throughout the UI
Persistence	Browser LocalStorage	Data persists across sessions without a backend
Hosting	GitHub Pages	Zero-cost, zero-config static web hosting

4. Unique Points

- **Authentic SAP MM Simulation:** Every page maps to a real SAP transaction code (MM01, XK01, ME21N, MIGO, MMBE) making this directly educational for SAP learners.
- **Complete Procurement Cycle:** The app enforces the real SAP procurement flow — only Approved POs can have GR posted, and posting a GR automatically updates live stock and closes the PO.
- **No Backend Required:** Entire application runs client-side. Data persists via LocalStorage, eliminating the need for any server, database, or API configuration.
- **SAP-Authentic UI Design:** Custom CSS design system with SAP's official blue (#0057b8), sidebar layout, badge system, and transaction-code styled page headers — visually matches SAP Fiori.
- **Real-time Stock Alerts:** Dashboard and stock page dynamically calculate and highlight materials below their reorder point with visual progress bars and color-coded status badges.
- **Multi-chart Analytics:** Reports page includes 6 different Chart.js visualizations covering vendor performance, stock valuation by group, top materials, and material type distribution.
- **GitHub Pages Ready:** Zero-dependency deployment — clone, push, enable Pages. Works immediately without npm install, build steps, or any configuration.

5. Project Structure

```
sap-inventory-system/  
index.html – Main dashboard  
README.md – GitHub documentation  
css/  
style.css – Complete design system (750+ lines)  
js/  
data.js – Shared data store + LocalStorage  
app.js – Utility functions (toast, format)  
dashboard.js – Dashboard charts and KPIs  
pages/  
materials.html – Material Master (MM60)  
vendors.html – Vendor Master (XK01)  
purchase-orders.html – Purchase Orders (ME21N)  
goods-receipt.html – Goods Receipt (MIGO)  
stock.html – Stock Overview (MMBE)  
reports.html – Analytics & Reports  
docs/  
Project_Documentation_...pdf – This document
```

6. Screenshots & UI Overview

The application features a fully responsive SAP Fiori-inspired interface. Below is a description of each module's UI as deployed in the browser:

Dashboard ([index.html](#))

4 KPI cards showing total materials, open POs, low stock alerts, and active vendors. Below are a dual-axis bar chart comparing POs vs Goods Receipts monthly, a doughnut chart for stock by material group, and two data tables for recent POs and low-stock materials.

Material Master ([materials.html](#))

Full data table with search, material type filter (ROH/HALB/FERT), edit and delete actions. Add Material modal includes fields for ID, name, type, UOM, plant, price, stock, reorder point, and group.

Vendor Master ([vendors.html](#))

Vendor list with star ratings, payment terms, city, status badges. Add/Edit modal with all vendor fields. Status filter between Active and Inactive vendors.

Purchase Orders ([purchase-orders.html](#))

Stats row showing total, pending, approved count and total value. Approve button converts Pending POs to Approved. Create PO modal with dynamic line item builder — select material, enter qty/price, add multiple items, calculates running total.

Goods Receipt ([goods-receipt.html](#))

Upper panel selects from Approved POs and previews line items with editable receipt quantities. Posting a GR updates stock in real-time, closes the PO, and adds entry to GR history table.

Stock Overview (stock.html)

Each row shows material stock with a visual progress bar relative to reorder point. Green/orange/red coloring indicates stock health. Adjust panel allows adding stock (goods issue) or reducing it (goods issue) with instant update.

Reports (reports.html)

4-tab report section: Vendor Analysis (bar + pie + summary table), Stock Valuation (bar + doughnut), PO Summary (stats + full table), Material Type (bar + plant distribution pie).

7. SAP Concepts Applied

Material Master (MM01/MM60)

Materials are categorized using standard SAP material types: ROH for raw materials, HALB for semi-finished goods, and FERT for finished products. Each material has a plant assignment, unit of measure, base price, and reorder point — exactly as defined in SAP MM configuration.

Vendor Master (XK01)

Vendors are managed with payment terms (Net 30/45/60), vendor IDs, contact details, and status flags — mirroring the vendor master structure used in SAP's Accounts Payable and MM modules.

Purchase Order (ME21N)

POs are created with header data (vendor, dates) and line items (material, quantity, price). The approval workflow simulates SAP's release strategy where POs move from Created to Approved before a GR can be posted.

Goods Receipt (MIGO - Movement Type 101)

GR posting is restricted to Approved POs only. On posting, the system updates unrestricted stock automatically — matching SAP MIGO movement type 101 (GR for Purchase Order).

Stock Management (MMBE/MB52)

The stock overview shows unrestricted-use stock per material per plant with real-time valuation. The reorder point monitoring mirrors SAP's MRP planning logic for identifying materials requiring replenishment.

8. Future Improvements

- SAP S/4HANA OData API Integration — Connect to real SAP backend via RESTful OData services for live data sync.
- Invoice Verification (MIRO) — Add three-way matching between PO, GR, and vendor invoice.
- Role-Based Access Control — Separate logins for MM Consultant, Purchase Manager, Store Keeper, and Finance user.
- MRP Run Simulation — Automatic reorder suggestion when stock drops below reorder point.
- Batch/Serial Number Tracking — Track individual material batches with expiry dates.
- Export to Excel/PDF — Allow users to download any report as a formatted Excel or PDF file.

- Email Notifications — Automated alerts for low stock, PO approval pending, and GR posting confirmation.
- Multi-Currency Support — Handle international vendors with currency conversion.
- Audit Trail / Change Log — Track all changes made to master data with timestamp and user ID.

Declaration

I hereby declare that this project is my own original work and has not been copied from any other student, senior, or internet source. All code has been written by me as part of the Capstone Project for the SAP Track program.

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